

Mobile APP Remote Control

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1. Scan QR Code to Install APP
2. Start the Robot
3. Connect APP to Robot
 - Select Device
 - Establish Network Connection
4. APP Function Introduction
 - Remote Control
 - Performance
 - FPV Control
 - Data Monitoring
 - Robot Calibration

1. Scan QR Code to Install APP

For Android users, please open the Google Play Store and search for "YahboomRobot", or open your mobile browser, scan the QR code below to download and install the "YahboomRobot" APP.

For iOS users, please open the App Store and search for "YahboomRobot", or use a QR code scanner to scan the QR code below to download and install the "YahboomRobot" APP.

If the latest version of the "YahboomRobot" APP is already installed on your phone, you do not need to install it again.



2. Start the Robot

The TF card provided with the product contains the image system by default. Insert the TF card into the motherboard's TF card slot, and after assembling the Muto robot, it can be turned on normally without reconfiguring the image.

Press the switch button at the tail of the Muto robot. The switch is a self-locking type, meaning it does not spring back after being pressed, and the robot remains continuously powered on, with the green light on the switch staying lit. The Muto robot will stand up. Wait for about 1.5 minutes for the system to start. After the system starts, the buzzer will beep three times. At this point, you can see the information displayed on the OLED screen.

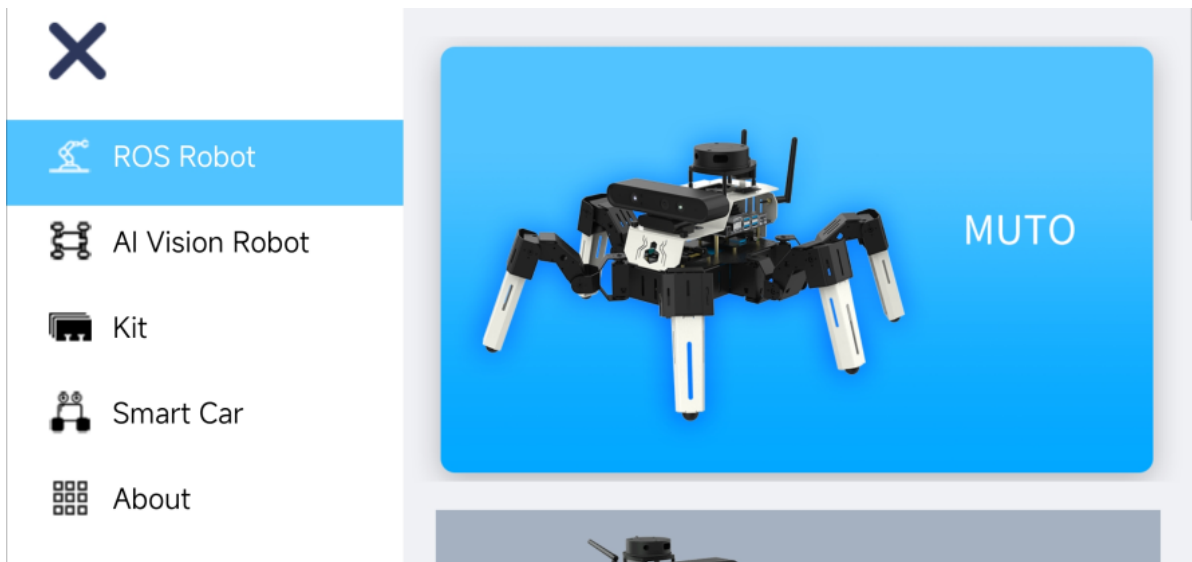
Raspberry Pi system username: pi password: yahboom

3. Connect APP to Robot

The factory system comes with a hotspot signal "**Muto_WIFI**" with the password "**12345678**". You can first connect your phone to the robot's hotspot to form a local area network. Alternatively, you can connect both the robot and your phone to the same router to form a local area network.

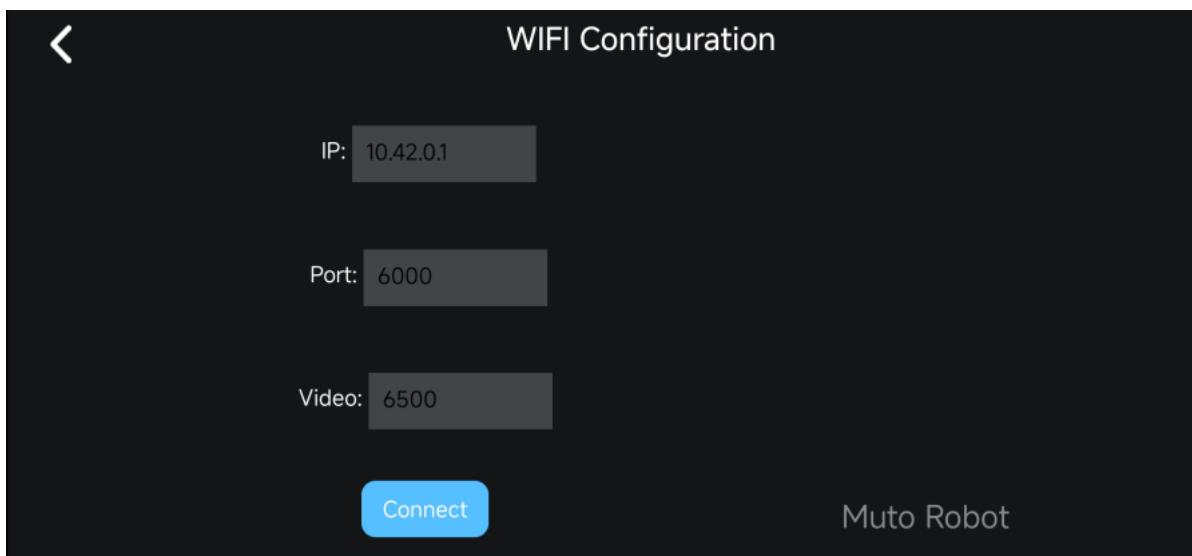
Select Device

When you first open the "YahboomRobot" APP, you need to select the "Muto" device under "ROS Robot" according to the model of the robot you purchased.



Establish Network Connection

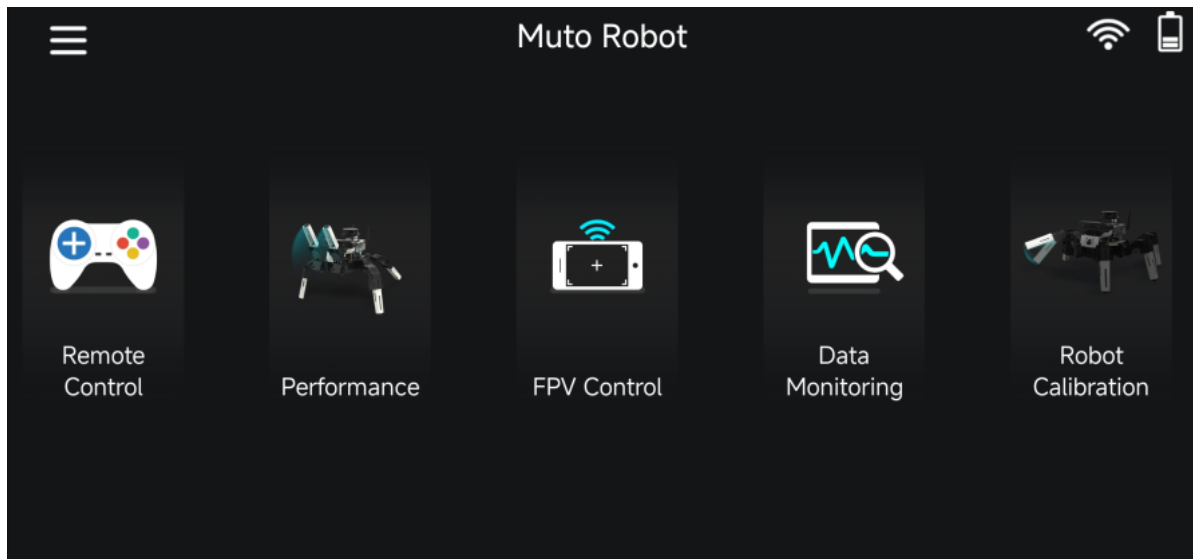
Enter the IP address displayed on the robot's OLED screen in the IP field, enter 6000 in the Port field, and 6500 in the Video field. Click "Connect". After a successful connection, it will automatically jump to the main control interface.



Note: Before connecting the device, please ensure that your phone is connected to the robot's hotspot, or that both the phone and the robot are connected to the same router. Also, make sure that the App control program has been started (the factory system starts the App control program by default on boot).

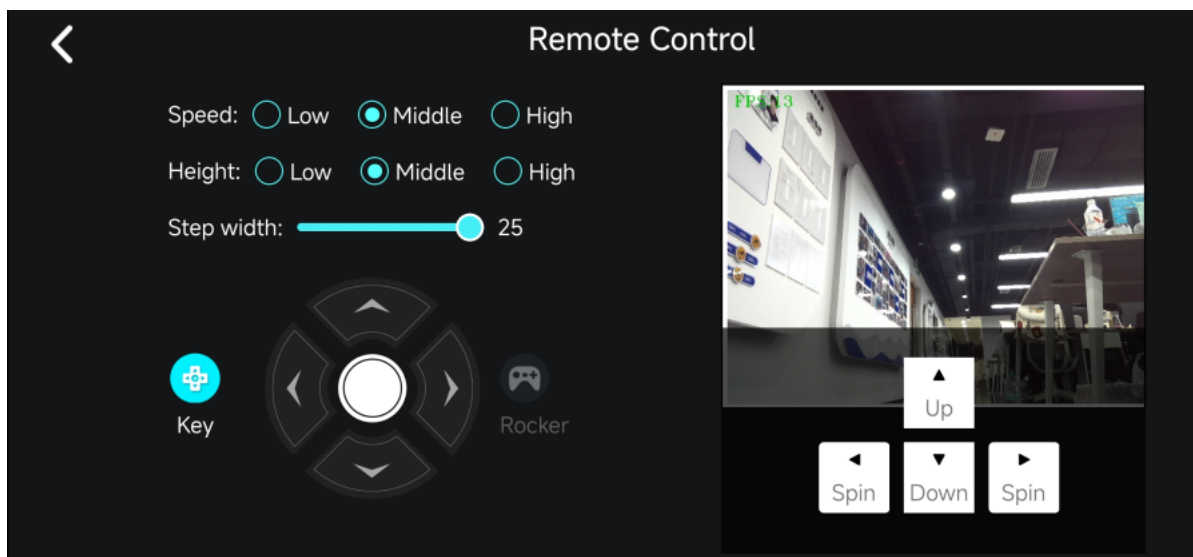
4. APP Function Introduction

The main interface of the Muto robot's APP is divided into five modules, each corresponding to different functions.



Remote Control

The function of the "Remote Control" interface is shown below.



Part 1. Speed: Controls the robot's movement speed. You can choose low, medium, or high speed.

Part 2. Height: Controls the robot's body height. You can choose low, medium, or high.

Part 3. Step Width: Controls the width of one step of the robot.

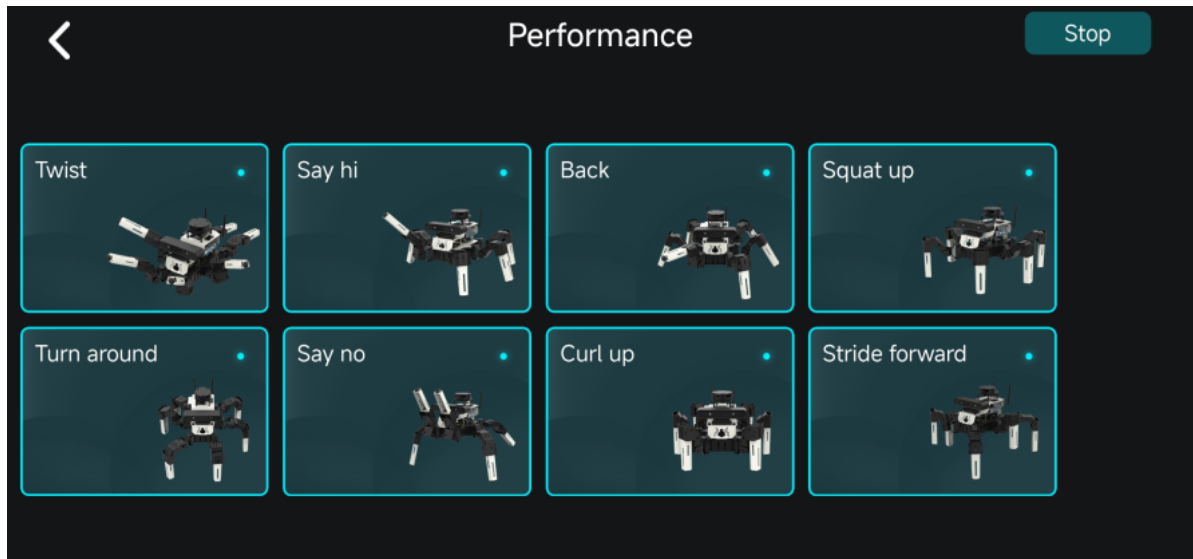
Part 4. Control Method: On the left, select button control: up for forward, down for backward, left for strafe left, right for strafe right, and the middle button to stop. On the right, select joystick control: up for forward, down for backward, left to turn left, right to turn right, and the center to stop.

Part 5. Up/Down: Controls the robot to look up/down.

Part 6. Left/Right Spin: Controls the robot to turn left/right in place.

Performance

The function of the "Performance" interface is shown below.



Part 1. Eight preset actions, each runs once per click.

Part 2. Reset: Resets the robot to its default posture.

FPV Control



Part 1. Hide: Hides the controls, keeping the full-screen camera view. Click again to show the controls.

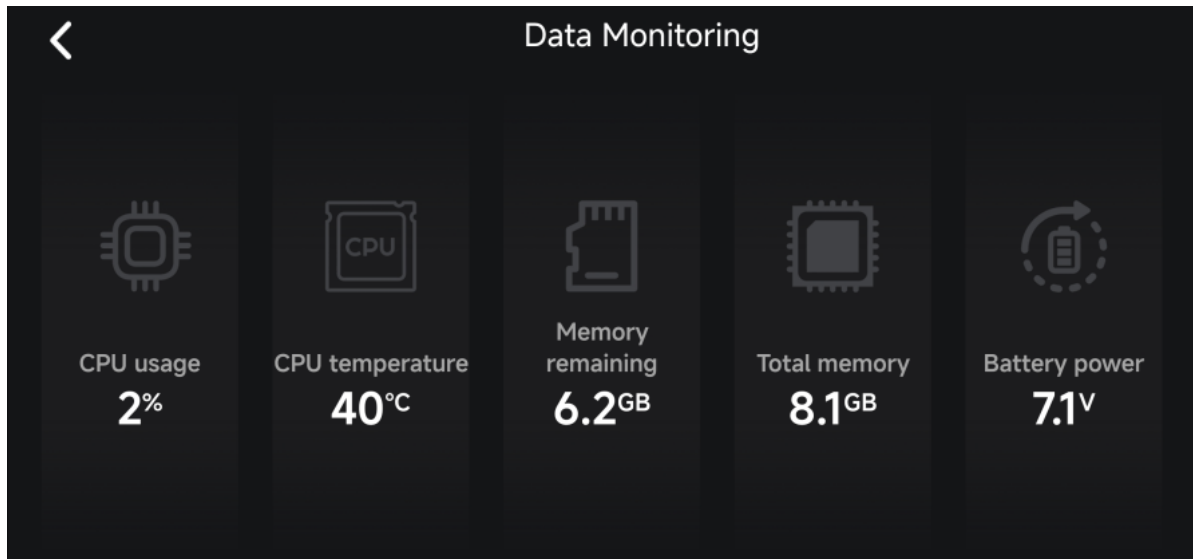
Part 2. Joystick: Controls the robot's movement.

Part 3. X/Y Axis Servos: Controls the rotation of the camera servo gimbal.

Note: The RS version lacks servos, so this function is not available.

Data Monitoring

The function of the "Data Monitoring" interface is shown below.



Part 1. CPU Usage: Displays the CPU usage of the mainboard.

Part 2. CPU Temperature: Displays the CPU temperature of the mainboard.

Part 3. Memory Remaining: Displays the amount of remaining system memory on the mainboard in GB.

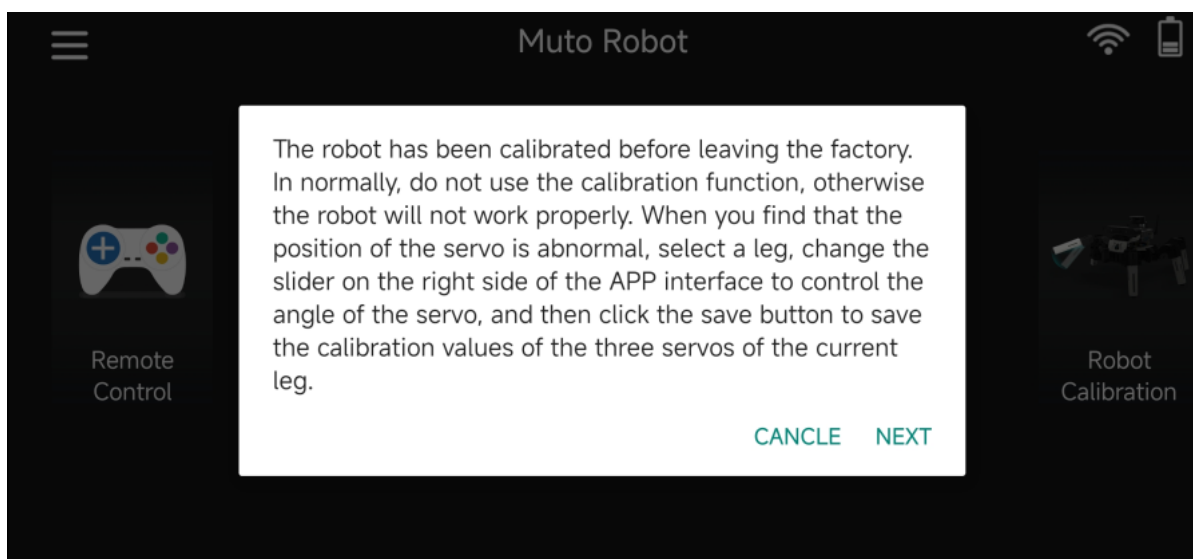
Part 4. Total Memory: Displays the total system memory on the mainboard in GB.

Part 5. Battery Power: Displays the robot's battery voltage in volts.

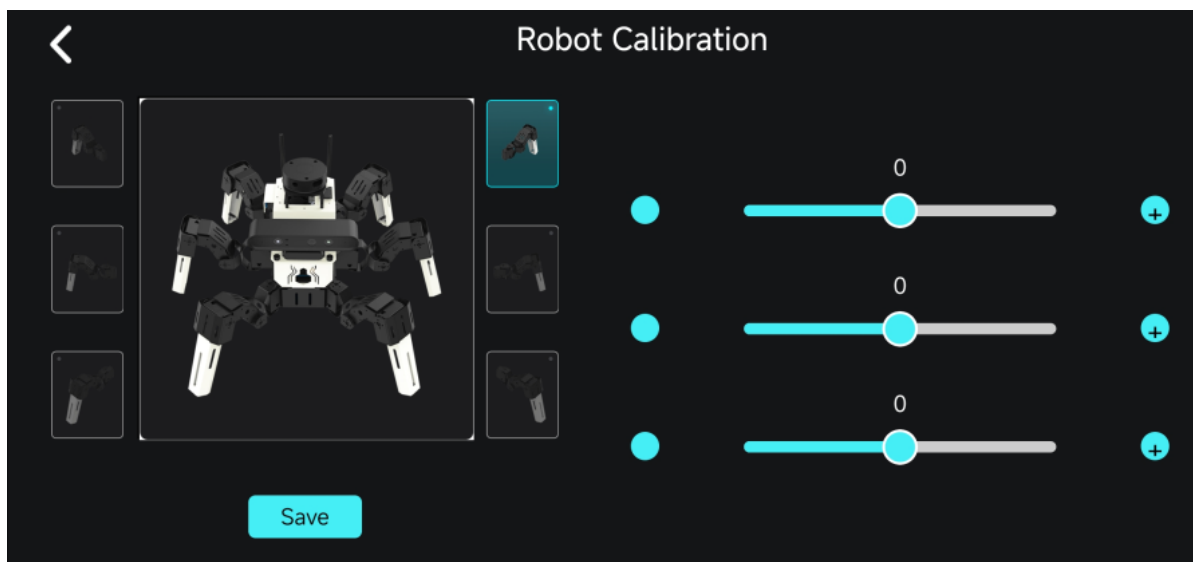
Robot Calibration

Note: The robot has been calibrated at the factory and does not need to be calibrated under normal circumstances. As the robot calibration function is special, improper operation may cause issues such as unsmooth robot movement.

Before entering the robot calibration interface, the APP will display the following message. Please click "Next".



The "Robot Calibration" interface is shown below.



As shown in the figure above, the image on the left represents the selected leg of the robot. If the icon is lit, it is selected. The save button at the bottom is used to save the data to the robot's baseboard. The three sliders on the right control the three servos of the selected leg.

Please place the Muto robot flat on the ground, facing you. If you see a leg suspended in the air, select the corresponding leg icon (if the icon is lit, it is selected, and the values of the three sliders on the right are all 0), and click the save button to clear the previously saved data. Then, adjust the three sliders on the right until the problematic leg touches the ground. Finally, click the save button again to save the data to the robot's servos.